

YASH THAKKAR

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Education

Princeton University

Expected May 2027

Bachelor of Science in Computer Science, Minors in Math and Robotics

Princeton, NJ

- **Relevant Coursework:** Distributed Systems, Reinforcement Learning, Algorithms & Data Structures, Machine Learning, Advanced Programming Techniques, Computer Vision, Robotics (Drone Lab), Probability, Operating Systems.
- **Extracurricular:** AI @ Princeton, HackPrinceton, TigerApps, Princeton ACM, Robotics Club, CodePath.

Experience

Google

May 2026 – Present

Software Engineer Intern

Mountain View, CA

- Built a **C++** service detecting orphaned **Spanner** databases on petabyte-scale Google Workspace recovery workflows.
- Designed state-diffing & TTL algorithms via Spanner management APIs, cutting storage overhead.
- Built real-time **Streamz** alerting with time-series monitoring, improving overall data-store reliability.

Princeton University Computer Science

September 2025 – Present

Head Teaching Assistant

Princeton, NJ

- Lead TA operations across three intro CS courses, supporting **300+ students per semester** in **Java** and **C**.
- Mentor students on **data structures**, **systems programming**, debugging code, and teach core CS concepts.
- Built TigerHelp, a **Next.js/PostgreSQL** office-hours platform with SSE live queue, CAS auth, & TA analytics view.

LinkedIn

May 2025 – August 2025

Software Engineer Intern

Mountain View, CA

- Engineered an AI recruiting agent via **MCP server**, automating 12 workflows and boosting recruiter task speed by 6x.
- Built AI Agent backend feature with Lucene search using **Python** and **Java**, boosting candidate eval relevance by 12%.
- Developed **TTL-based termination system** for large-scale LLM workflows, cutting wasted compute by 30%.

Fidelity Investments

June 2024 – August 2024

Software Engineer Intern

Durham, NC

- Developed a web app using **JavaScript** to analyze cloud workflow portability, enhancing decisions for 25+ teams.
- Integrated **Claude 3.5 LLM** with **AWS Bedrock** to automate architecture reviews, improving efficiency by 40%.
- Engineered a comprehensive knowledge base for embedding and RAG, increasing model accuracy by 75%.

NoTraffic

June 2023 – August 2023

Software Engineer Intern

Tel Aviv, Israel

- Created an AI-based accident detection MVP using **Django-Angular**, deployed to 150+ traffic intersections.
- Devised a **Python** script with **ML** and **OpenCV** to sync 4 video streams, increasing accident detection by 35%.

Technical Skills

Languages: Python, C++, JavaScript, Java, C, Swift, HTML/CSS, SQL, R, Assembly, PostgreSQL, MySQL.

Libraries/Frameworks: PyTorch, TensorFlow, Scikit-learn, FastMCP, React, OpenAI, OpenCV, Flask, Node.js, Kafka.

Developer Tools: CUDA, Git, MLFlow, Claude Code, Docker, Kubernetes, AWS, CI/CD, Spanner.

Research

Multi-Agent RL for Imperfect-Information Games | *Python, PyTorch* | [GitHub](#) | [Paper](#)

Spring 2026

- Built a multi-agent **self-play** loop with **PPO**, action masking, and belief-state features for partial observability.
- Surpassed experienced human players, developing complex, emergent bidding strategies directly from raw game states.

A Smarter Search for Tree of Thoughts | *Python, GPT-4 API* | [GitHub](#) | [Paper](#)

Spring 2026

- Co-authored paper replacing ToT's BFS/DFS with **A*** (lazy LLM-scored heuristic) and **MCTS** with UCT exploration.
- Found evaluator calibration bounds search gains; MCTS beat BFS only when state evaluations were reliable.

Semantic Reasoning Evaluation in LLMs | *PyTorch, Qwen2.5-Coder-32B, HuggingFace* | [GitHub](#) | [Paper](#)

Fall 2025

- Generated **synthetic datasets** to evaluate **Qwen's** semantic reasoning via obfuscation and error-detection ablations.
- Implemented an **LLM-as-a-judge**, proving the model preserves intent but fails at **multi-step error localization**.

Projects

Low-Cost Drone Delivery Platform | *C++, Arduino, CAD* | [GitHub](#) | [Paper](#)

Spring 2026

- Designed an open-source autonomous payload-drop system (ultrasonic ground detection, 3D-printed mechanisms); iterated across 7 designs to reach **84% bench** / **80% flight** release reliability.

Pedestrian Detection Benchmark | *Python, PyTorch* | [GitHub](#) | [Paper](#)

Fall 2025

- Benchmarked **HOG**, **CNN**, **ViT**, and **SAM2** pipelines on the PnPLO pedestrian dataset, comparing accuracy-latency tradeoffs across classical and foundation-model approaches.